

The AMD Slingshot logo is displayed in white. It features the word "AMD" in a bold, sans-serif font, followed by a square icon containing a stylized, geometric arrow pointing to the right. Below this, the word "Slingshot" is written in a similar sans-serif font.

AMD
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HUMAN *IMAGINATION*
BUILT WITH *AI*

Powered by **I I2S**

Team Details

Team name - **Fang**

Team leader name - **Md Sadique Imam**

Problem Statement - **Autonomous Temporal
Semantic Archive (LIFELOG AI)**

Brief about the idea

A small camera + microphone runs in the background 24/7 → captures 10-second video+audio pulses → automatically extracts objects (with exact positions & movements), environment context, spoken words, and their meaning → stores everything in a time-ordered, semantically indexed archive that you can later query naturally (e.g. “where is my black pen?”, “what did I promise Sarah yesterday?”, “show me every time I touched the charger today”).

Key properties

Autonomous = zero user effort (no “hey assistant”, no buttons, no tagging)

Temporal = everything is timestamped and sequenced (you can reconstruct chronology)

Semantic = not just raw video/audio — the system understands meaning (object identity, spatial relations, intent behind words, commitments, actions)

Archive = persistent, queryable long-term storage (logs which will be search using RAG)

Opportunities

- How different is it from any of the other existing ideas?
- How will it be able to solve the problem?
- USP of the proposed solution

→ Unlike audio-only tools (Rewind.ai, Limitless) that lack visual context, photo-only lifeloggers (SenseCam) without audio/semantics, or screen-bound logging (Microsoft Recall), this system passively fuses dense video—objects, exact positions, movements—with verbatim speech and intent extraction in a zero-input way, capturing the physical world objectively.

→ It eliminates manual effort and memory gaps by continuously indexing every moment (objects, locations, commitments). Queries later retrieve precise, visual-backed answers with context and chronology.

→ Unlike reactive chatbots (**ChatGPT**, **Grok**) that depend on user prompts and web data, it silently observes real-time reality, resolves ambiguities (“this” → held object), and delivers objective, effortless, private recall of daily life—turning fragmented memory into reliable, searchable reconstruction.

List of features offered by the solution

Fully passive, zero-input capture — No wake words, buttons, or manual triggers; runs continuously in the background.

High-frequency multi-modal recording — 10-second video-audio pulses at up to 10 fps from a single camera + microphone.

Dense object detection & tracking — Identifies every visible object with attributes (color, type, material) and precise spatial positions relative to fixed anchors (e.g., “black pen on desk left of laptop”).

Full environment awareness — Captures room type, lighting, walls, floor, furniture, and overall atmosphere in every pulse.

Verbatim speech transcription with timestamps — Accurate audio-to-text with time codes for each spoken segment.

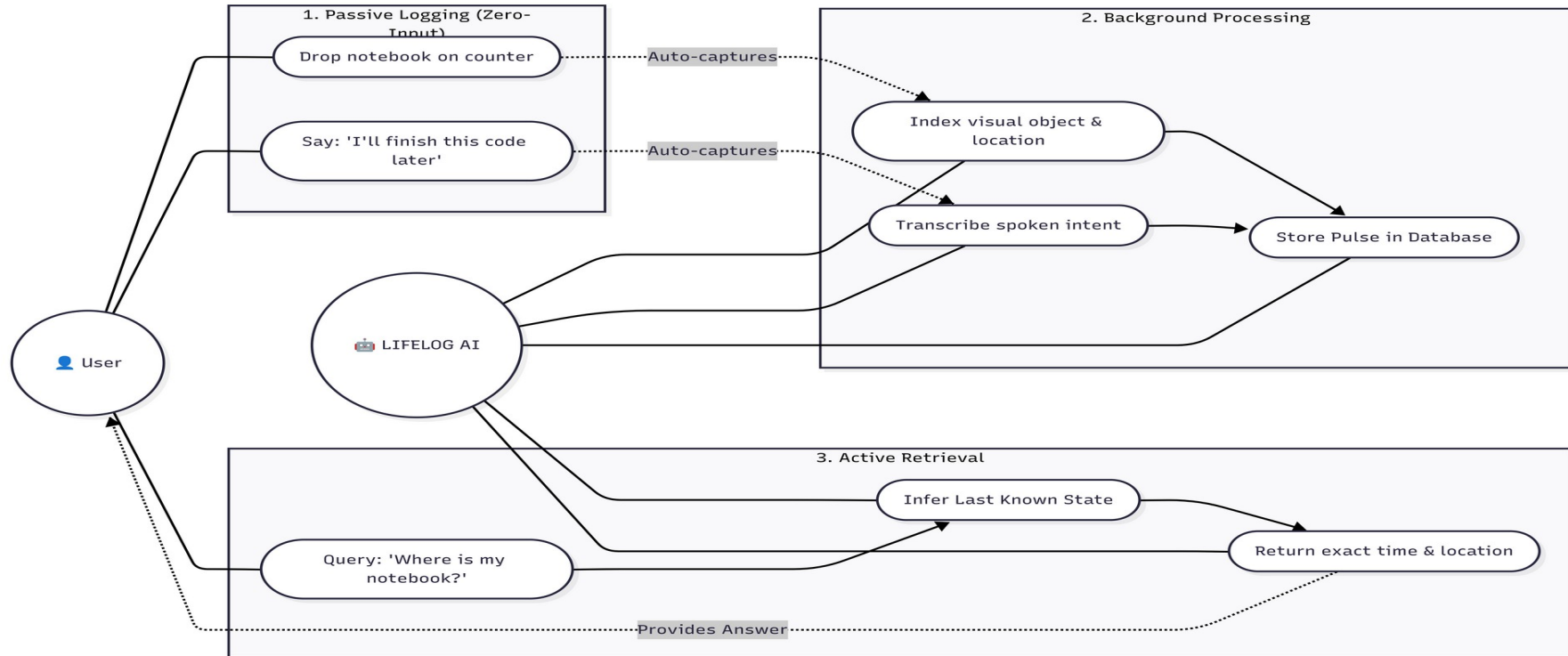
Semantic intent & commitment extraction — Understands references (“this”, “my keys”), plans, promises, and context from speech.

Time-ordered semantic indexing — All data stored chronologically with timestamps, enabling precise temporal queries.

Natural-language querying — Retrieve moments via questions like “Where did I put my black pen?”, “What did I say about the meeting?”.

Lightweight & scalable architecture — Runs on consumer hardware (e.g., single T4 GPU or edge device) with two-stage LLM pipeline for efficiency.

Use-case diagram

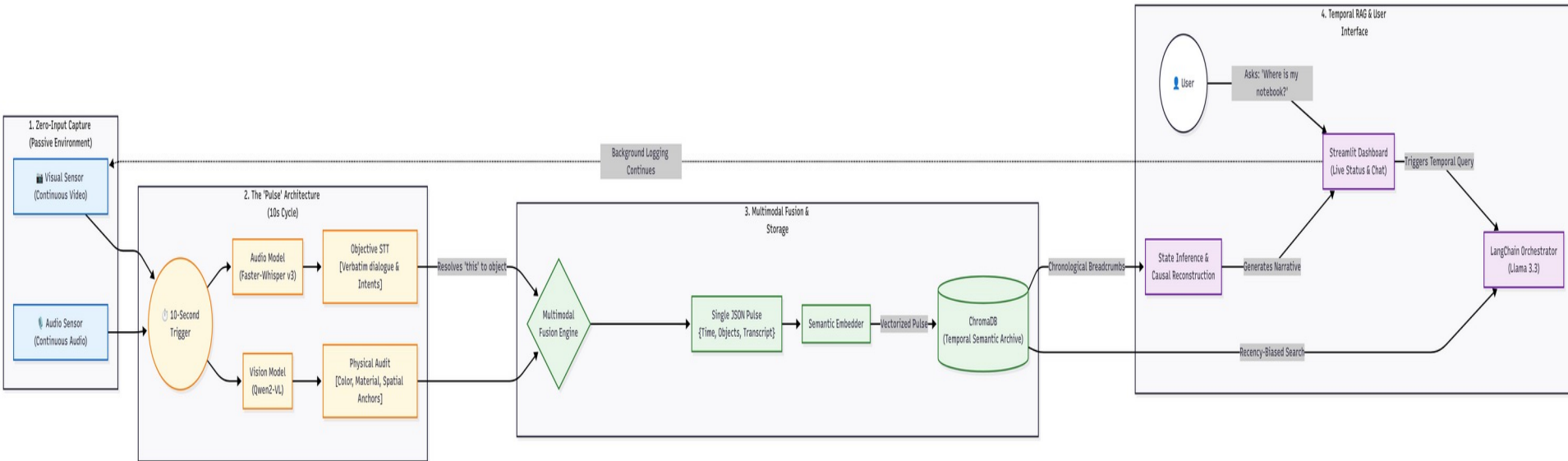


Wireframes/Mock diagrams of the proposed solution (optional)



Sample Home Page

Architecture diagram of the proposed solution



Technologies to be used in the solution

Vision Model: Qwen2-3B_VL for native video-language reasoning and attribute-dense object indexing.

Audio Intelligence: Faster-Whisper v3 for objective Speech-to-Text and isolation of verbatim commitments.

Vector Database: ChromaDB to store multimodal "Pulses" for recency-biased semantic search.

Reasoning Engine: Gemini/Grok LLM (API) for analyzing temporal breadcrumbs and generating causal narratives.

Orchestration: LangChain to manage the Temporal RAG pipeline between the database and LLM.

Frontend: Streamlit for a live dashboard showing room status and historical chat.

Usage of AMD Products/Solutions: AMD Ryzen™ AI 400 Series

Tri-Engine Workload Optimization

We utilize the **AMD XDNA™ 2** architecture to distribute the "Pulse" workload across three specialized engines:

NPU (Neural Processing Unit): We offload continuous, background tasks to the NPU (delivering up to 60 TOPS). It handles Faster-Whisper for real-time speech-to-text and supports 16K context lengths, allowing the AI to "remember" hours of conversation while maintaining multi-day battery life.

iGPU (Integrated GPU): The RDNA 3.5 Graphics accelerates our Vision-Language Model (Qwen2-3B_VL). This enables "Attribute-Dense Vision" for high-fidelity object identification and spatial mapping without slowing down the rest of your system.

CPU (Zen 5): The high-performance cores manage the ChromaDB vector lookups, LangChain orchestration, and the Streamlit dashboard, ensuring an instant, snappy user experience.

The AMD Advantage

Privacy-First: Using the Ryzen AI Software , all multimodal processing stays on-chip. Your physical and verbal logs never leave your device, protected by hardware-level AMD PRO security.

Local Efficiency: Through Vitis AI Quantization , we achieve 2x lower latency and a reduced memory footprint, allowing it to run smoothly on standard 16GB RAM systems.

Always-On Utility: The NPU's extreme efficiency allows the system to index your world 24/7 in a "Passive Auditor" state without generating excess heat or fan noise.

Estimated implementation cost : Comparison - Traditional VS AMD Version

Cost Category	Traditional Cloud Stack	LIFELOG AI (AMD Ryzen AI)
Hardware	\$1,000 (Basic Laptop)	\$2,200 (AI PC)
Inference Fees	\$200 - \$500 / month	\$0 (Local)
Storage Fees	\$50 - \$150 / month	\$0 (Local)
1-Year Total	\$4,000 - \$8,800+	\$2,200 (One-time)

Prototype Assets (Optional)

GitHub Public Repository Link: https://github.com/FANGxPC/LIFELOG_AI

→ Demo Video Link (Max: 3 Minutes): **Not Available YET**

Add as per the requirements of the contest



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Thank you!

